

COMP6237 Data Mining

Lecture 5: Recommendation Systems

Zhiwu Huang

Zhiwu.Huang@soton.ac.uk

Lecturer (Assistant Professor) @ VLC of ECS
University of Southampton

Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

[Book time with Zhiwu Huang: Office Hour](#)

(Thanks to Prof. Jonathon Hare and Dr. Jo Grundy for providing the lecture materials used to develop the slides.)

Recommendation Systems – Problem



	Life is Beautiful	Seven Samurai	Joker	Schindler's List	The Pianist	City Of God
Lee	2.5	3.5	3.0	3.5	3.0	2.5
Sofia	3.0	3.5	1.5	5.0	3.0	3.5
Miley	2.5	3.0	NaN	3.5	4.0	NaN
Justina	NaN	3.5	3.0	4.0	4.5	3.0
Donald	3.0	4.0	2.0	3.0	3.0	2.0
Mickey	3.0	4.0	NaN	5.0	3.0	3.5
Tristan	NaN	4.5	NaN	4.0	NaN	1.0

Context: This table is a mini “Netflix”. We have 7 users, 6 films, and some of their 1–5 star ratings. Many entries are missing – we don’t yet know how every user feels about every film.

Recommendation Systems – Problem

Problem: The user-item rating table is **incomplete**. Using only the observed ratings, can we **predict the missing entries** — in particular **Miley's ratings** for the two unseen films “**Joker**” and “**City Of God**”? In other words: what should we **recommend to Miley next**?

	Life is Beautiful	Seven Samurai	Joker	Schindler's List	The Pianist	City Of God
Lee	2.5	3.5	3.0	3.5	3.0	2.5
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Miley	2.5	3.0	NaN	3.5	4.0	NaN
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Donald	3.0	4.0	2.0	3.0	3.0	2.0
Mickey	3.0	4.0	NaN	5.0	3.0	3.5
Tristan	NaN	4.5	NaN	4.0	NaN	1.0

Today we'll focus on Miley: predict her missing ratings for **Joker** and **City of God**, then recommend the ones with high predicted ratings.

Recommendation Systems– Content-based Approach



	Life is Beautiful	Seven Samurai	Joker	Schindler's List	The Pianist	City Of God
Miley	2.5	3.0	NaN	3.5	4.0	NaN
romance	0.9	0.2	0.1	0.4	0.5	0.2
action	0.2	0.9	0.8	0.4	0.3	0.9

Content-based approach recommend items using item features. Here each film has two features: **romance** and **action** (between 0 and 1). The system learns Miley's **taste profile** from her rated films, then recommends new films whose (romance, action) features are closest to that profile.

Recommendation Systems– Content-based Approach

		Life is Beautiful	Seven Samurai	Joker	Schindler's List	The Pianist	City Of God
y	Miley	2.5	3.0	NaN	3.5	4.0	NaN
x_1	romance	0.9	0.2	0.1	0.4	0.5	0.2
x_2	action	0.2	0.9	0.8	0.4	0.3	0.9

Regression model: $y = f(X; \theta)$

Each film can be described by the vector $X = [1 \quad x_1 \quad x_2]$ (1 is for the bias term)

Learn 3D parameter vector θ , where $\theta^T X$ gives the number of stars for each user.

$\theta = [0 \quad 1 \quad 0]$ for someone who really likes romance films

$X = \begin{bmatrix} 1 & x_1 & x_2 \end{bmatrix}$

Recommendation Systems– Content-based Approach

		Life is Beautiful	Seven Samurai	Joker	Schindler's List	The Pianist	City Of God
y	Miley	2.5	3.0	NaN	3.5	4.0	NaN
x_1	romance	0.9	0.2	0.1	0.4	0.5	0.2
x_2	action	0.2	0.9	0.8	0.4	0.3	0.9

Regression model: $y = f(X; \theta)$

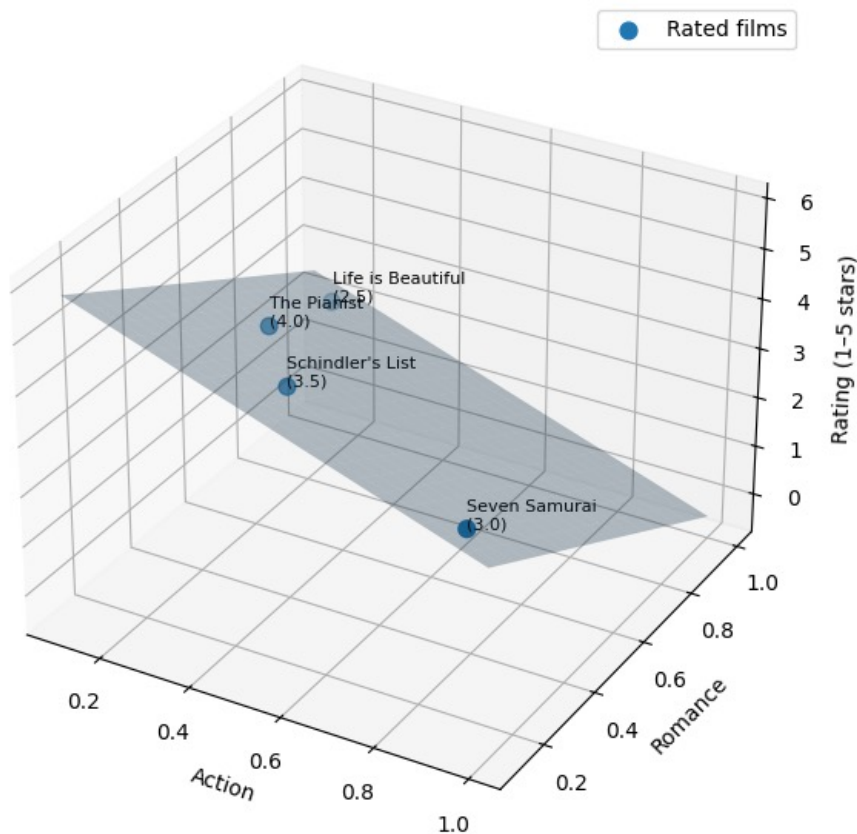
Use **Linear Regression** to find user parameter vector θ where m is the number of films rated by that user

$$\min_{\theta} \frac{1}{m} \sum_{i=1}^m (\theta^T X_i - y_i)^2 \quad (1)$$

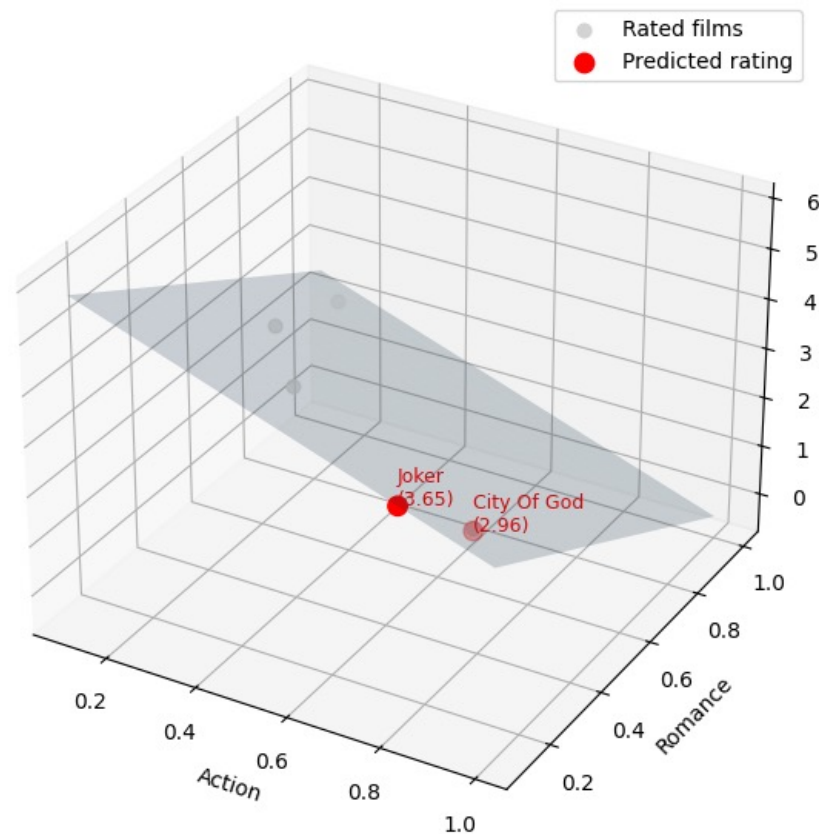
Can also use Bayesian classifiers, MLPs, etc.

Recommendation Systems– Content-based Approach

Miley's ratings and fitted regression plane



Miley: predicted ratings for Joker & City Of God



$\theta^T X$ predicts the ratings

Demo: Content-based recommendation

Recommendation Systems – Collaborative Approach

	Life is Beautiful	Seven Samurai	Joker	Schindler's List	The Pianist	City Of God
Lee	2.5	3.5	3.0	3.5	3.0	2.5
Sofia	3.0	3.5	1.5	5.0	3.0	3.5
Miley	2.5	3.0	NaN	3.5	4.0	NaN
Justina	NaN	3.5	3.0	4.0	4.5	3.0
Donald	3.0	4.0	2.0	3.0	3.0	2.0
Mickey	3.0	4.0	NaN	5.0	3.0	3.5
Tristan	NaN	4.5	NaN	4.0	NaN	1.0

Collaborative (user-based) systems use ratings from similar users. We find users whose rating patterns look like **Miley's** (e.g. Lee or Sofia). If they liked a film Miley hasn't rated yet, such as "**Joker**" or "**City Of God**", we predict Miley will like it too and recommend it.

Recommendation Systems – Collaborative Filtering

Collaborative filtering uses a range of approaches to accomplish this task

- ▶ Neighbourhood based approach
- ▶ Model based approach
- ▶ Hybrid (Neighbourhood and model) based approach

This lecture will cover the Neighborhood based approach

Recommendation Systems – Collaborative Filtering

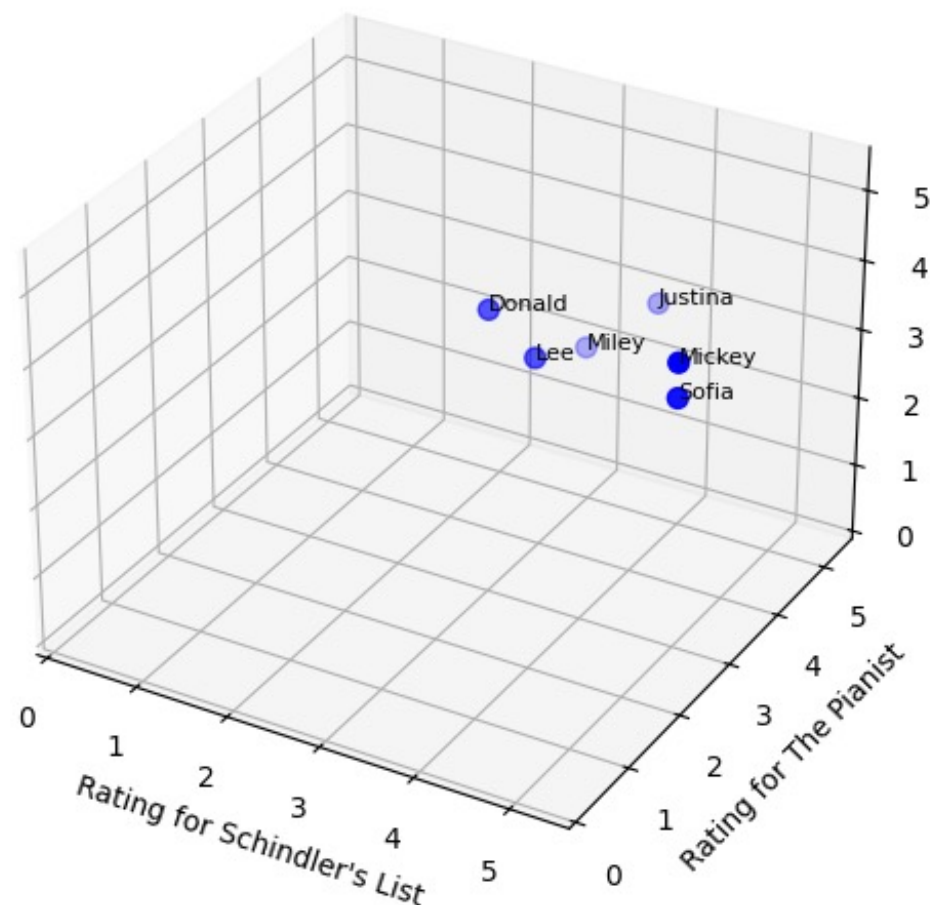
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Miley	2.5	3.0	NaN	3.5	4.0	NaN
Justina	NaN	3.5	3.0	4.0	4.5	3.0
Donald	3.0	4.0	2.0	3.0	3.0	2.0
Mickey	3.0	4.0	NaN	5.0	3.0	3.5
Tristan	NaN	4.5	NaN	4.0	NaN	1.0

Question: When can we say two users are “neighbours”?
What do we need to define before that statement even makes sense?

Collaborative Filtering – Feature

	Seven Samurai	Schindler's List	The Pianist
Lee	3.5	3.5	3.0
Sofia	3.5	5.0	3.0
Miley	3.0	3.5	4.0
Justina	3.5	4.0	4.5
Donald	4.0	3.0	3.0
Mickey	4.0	5.0	3.0
Tristan	4.5	4.0	NaN

Users in 3D movie-rating space

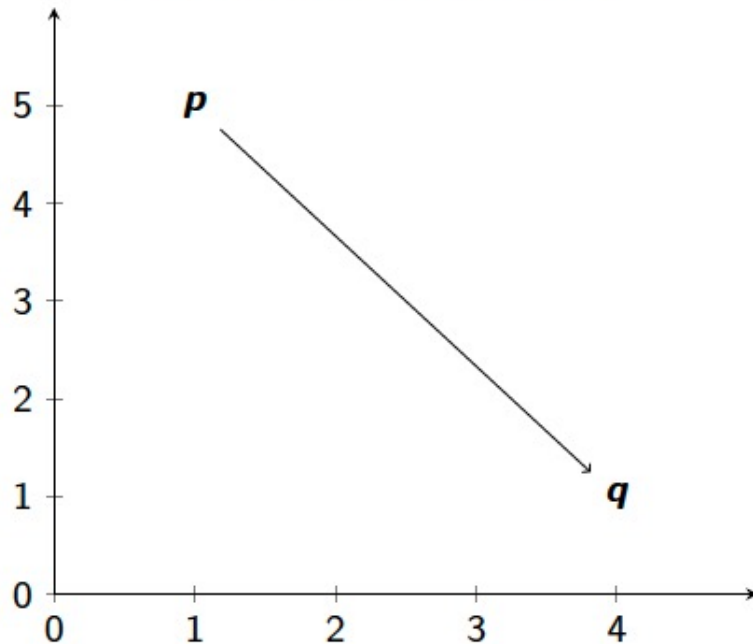


Question: where is Tristan in this 3D space?

Collaborative Filtering – Distance

There are a number of ways to measure distance in feature space:

► Euclidean Distance:



$\mathbf{p}$ and $\mathbf{q}$ are N-dimensional feature vectors,

$$\mathbf{p} = [p_1, p_2, \dots, p_N],$$

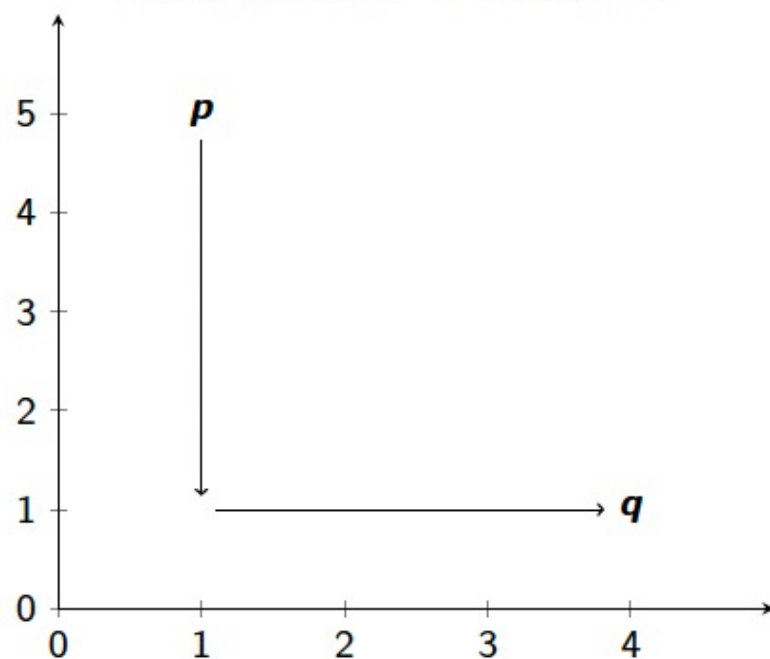
$$\mathbf{q} = [q_1, q_2, \dots, q_N]$$

Euclidean distance:

$$\|\mathbf{p} - \mathbf{q}\| = \sqrt{\sum_{i=1}^N (q_i - p_i)^2}$$

Collaborative Filtering – Distance

► Manhattan Distance:



$\mathbf{p}$ and $\mathbf{q}$ are N-dimensional feature vectors,

$$\mathbf{p} = [p_1, p_2, \dots, p_N],$$

$$\mathbf{q} = [q_1, q_2, \dots, q_N]$$

Manhattan distance:

$$\|\mathbf{p} - \mathbf{q}\|_1 = \sum_{i=1}^N |q_i - p_i|$$

Collaborative Filtering – User Similarity

There are many ways to compute similarity based on Euclidean distance

We *could* chose:

$$sim_{L2}(x, y) = \frac{1}{1 + \sqrt{\sum_{i \in I_{xy}} (r_{x,i} - r_{y,i})^2}}$$

where $r_{x,i}$ is the rating from user x for item i

I_{xy} is set of items rated by both x and y

i.e. when the distance is 0, the similarity is 1, but when the distance is large, similarity $\rightarrow 0$

Collaborative Filtering – User Similarity

Can also use cosine similarity

$$sim_{cos}(x, y) = \frac{\sum_{i \in I_{xy}} r_{x,i} r_{y,i}}{\sqrt{\sum_{i \in I_{xy}} r_{xi}^2} \sqrt{\sum_{i \in I_{xy}} r_{yi}^2}}$$

Only performed over the items which are rated by both users

Collaborative Filtering – User Similarity

Alternatively, calculate correlation of users, based on ratings they share

Using Pearson's Correlation: standard measure of dependence between two related variables.

$$\text{sim}_{\text{Pearson}}(x, y) = \frac{\sum_{i \in I_{xy}} (r_{x,i} - \bar{r}_x)(r_{y,i} - \bar{r}_y)}{\sqrt{\sum_{i \in I_{xy}} (r_{x,i} - \bar{r}_x)^2 \sum_{i \in I_{xy}} (r_{y,i} - \bar{r}_y)^2}}$$

Where $\bar{r}_x$ is average rating user x gave for all items in I_{xy}

Collaborative Filtering – User Filtering

We now have a set of measures for computing the similarity between users

Produce a ranked list of best matches to a target user. Typically want the top- N users

May only want to consider a subset of users, i.e. those who rated a particular item.

Collaborative Filtering – User-based Recommending

Now we have a list of similar users, how can we recommend items?

Predict rating $r_{u,i}$ of item i by user u as an aggregation of the ratings of item i by users similar to u

$$r_{u,i} = \text{aggr}_{\hat{u} \in U}(r_{\hat{u},i})$$

Where U is the set of *top* users most similar to u that rated item i

Multiply the score by the similarity of the user

Normalise by sum of similarities (otherwise items rated more often will dominate)

$$r_{u,i} = \frac{\sum_{\hat{u} \in U} \text{sim}(u, \hat{u}) r_{\hat{u},i}}{\sum_{\hat{u} \in U} |\text{sim}(u, \hat{u})|}$$

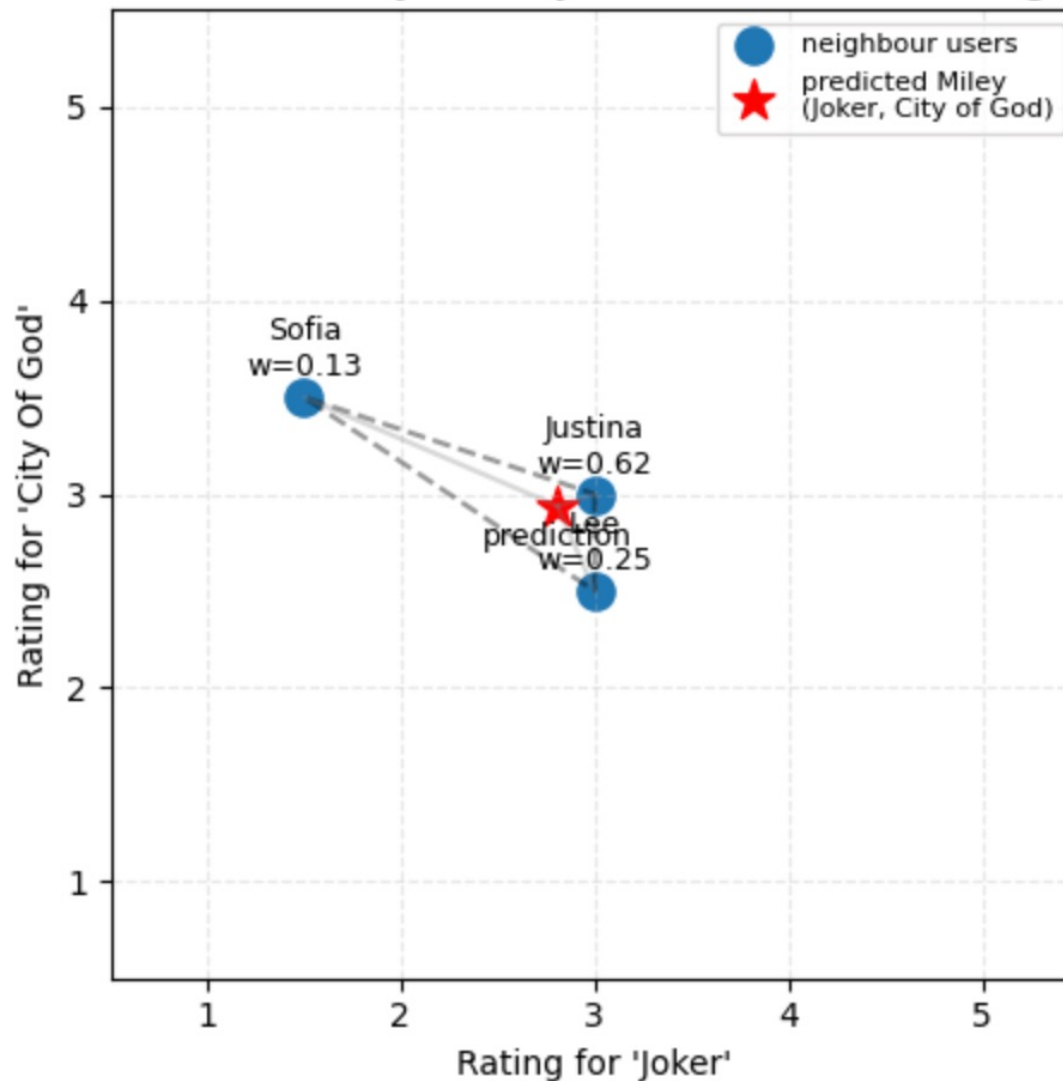
This is *User Based Filtering*

Target user: Miley

Neighbours (used for barycentric combination):

Justina	sim=1.000,	Joker=3.0,	CityOfGod=3.0
Lee	sim=0.405,	Joker=3.0,	CityOfGod=2.5
Sofia	sim=0.205,	Joker=1.5,	CityOfGod=3.5

User-based CF: Miley as barycentric combo of 3 neighbours



Barycentric prediction for Miley:

Joker:	2.81
City Of God:	2.94

Demo: User-based recommendation

Collaborative Filtering – Item-based Approach

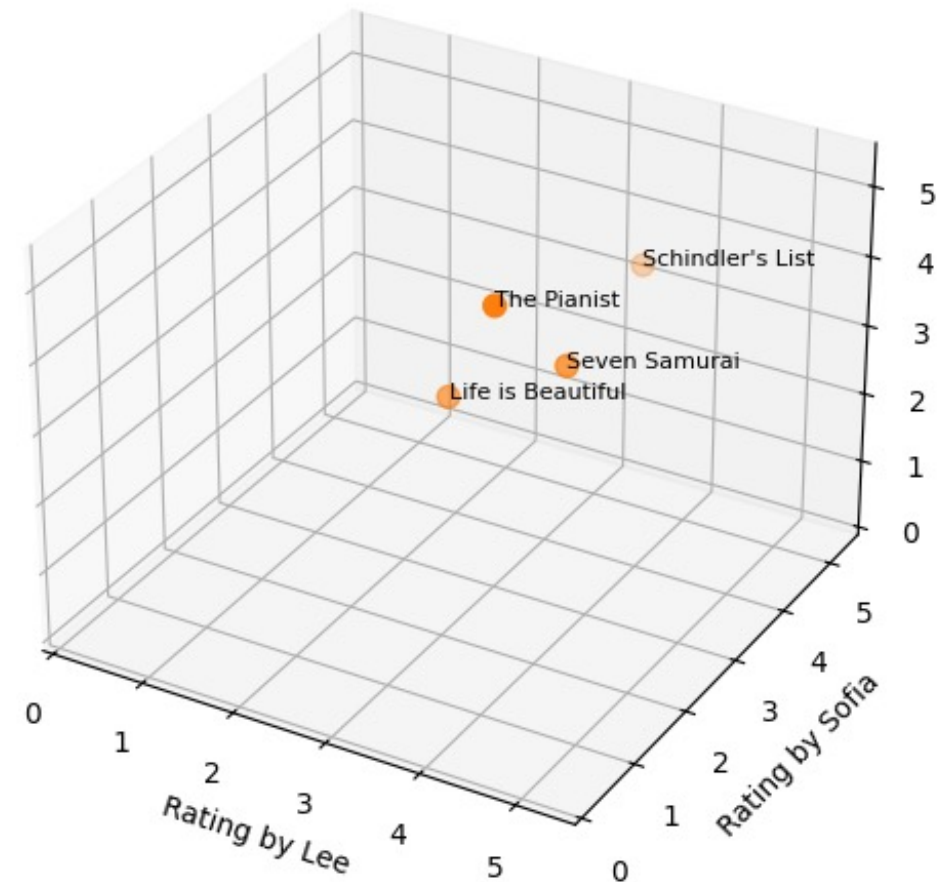
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Donald	3.0	4.0	2.0	3.0	3.0	2.0
Mickey	3.0	4.0	NaN	5.0	3.0	3.5
Tristan	NaN	4.5	NaN	4.0	NaN	1.0

Collaborative (item-based) systems use similarities between items. We look for films whose rating patterns are similar to ones Miley already likes. If “**Joker**” or “**City Of God**” look similar to films she rated highly, the system predicts she’ll like them too and recommends them.

Collaborative Filtering – Item-based Approach

Films in 3D rating space (item view)

	Seven Samurai	Schindler's List	The Pianist
Lee	3.5	3.5	3.0
Sofia	3.5	5.0	3.0
Miley	3.0	3.5	4.0
Justina	3.5	4.0	4.5
Donald	4.0	3.0	3.0
Mickey	4.0	5.0	3.0
Tristan	4.5	4.0	NaN



(Note: user-based CF: films as coordinates, and item-based CF: users as coordinates)

Collaborative Filtering – Item-based Approach

To compute recommendations using this approach:

Estimate the rating for unrated item $\hat{i}$ that has a top-N similarity to a rated item i :

$$r_{u,\hat{i}} = \frac{\sum_{i \in I} \text{sim}(\hat{i}, i) r_{u,i}}{\sum_{i \in I} \text{sim}(\hat{i}, i)}$$

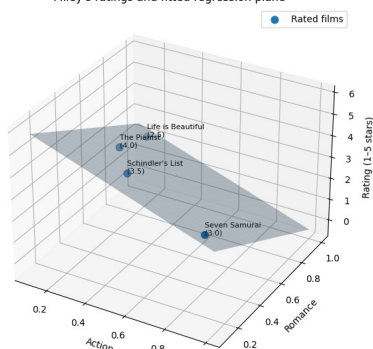
Where I is the subset of all N items similar to $\hat{i}$

[Demo: Item-based recommendation](#)

Recommendation Systems – Summary

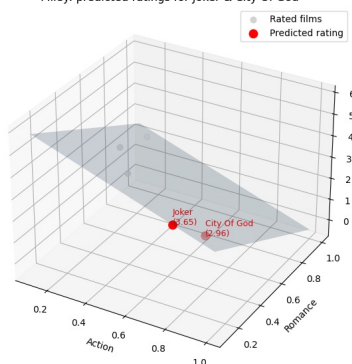
Content-based

Miley's ratings and fitted regression plane

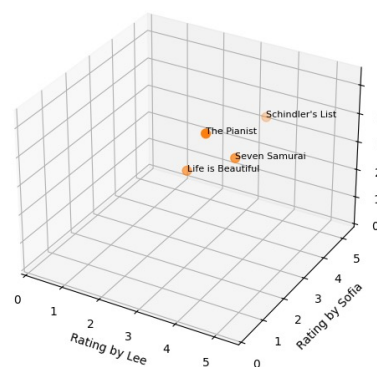


Users in 3D movie-rating space

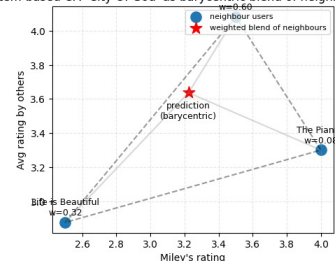
Miley: predicted ratings for Joker & City Of God



Films in 3D rating space (item view)

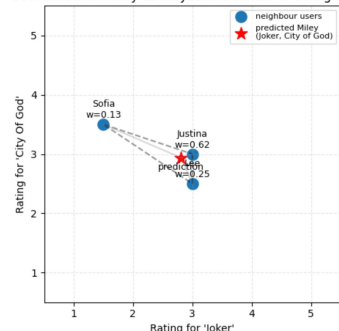


Item-based CF: 'City Of God' as barycentric blend of neighbour films



(Item-based)

User-based CF: Miley as barycentric combo of 3 neighb



User-based

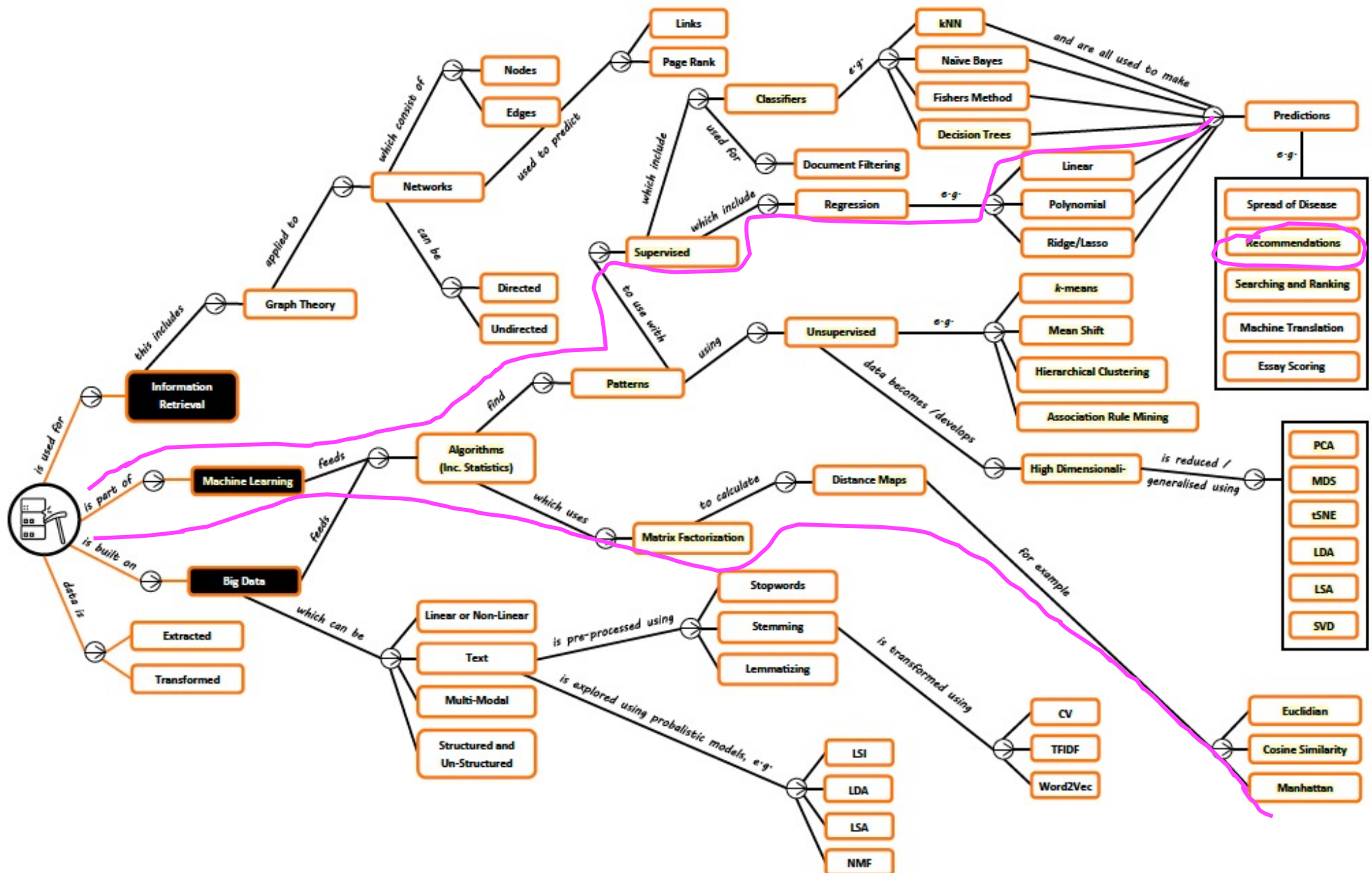
Approach

Big win

Main pain point

Content-based	Works for new items, few users	Needs good features, "more of same"
User-based CF	Very intuitive, serendipitous finds	Needs overlap, new users are hard
Item-based CF	Fast, widely used in real systems	New items are hard, needs many ratings

Recommendation Systems – Roadmap



Recommendation Systems – Textbook

Chapter 9

Recommendation Systems

There is an extensive class of Web applications that involve predicting user responses to options. Such a facility is called a *recommendation system*. We shall begin this chapter with a survey of the most important examples of these systems. However, to bring the problem into focus, two good examples of recommendation systems are:

1. Offering news articles to on-line newspaper readers, based on a prediction of reader interests.
2. Offering customers of an on-line retailer suggestions about what they might like to buy, based on their past history of purchases and/or product searches.

► Mining of Massive Datasets J. Leskovec *et al*

Recommendation Systems – Learning Outcomes

- **LO1:** Mastering fundamental concepts and mathematical calculations of content-based and user-based/item-based collaborative filtering approaches, such as (exam)
 - ❖ Measuring distances/similarities between users or items
 - ❖ Predicting the missing rating using content-based approach
 - ❖ Calculating the predicted rating with user-based/item-based collaborative filtering approach
- **LO2:** Implement basic algorithms using Python (coursework)

Assessment hints: Multi-choice Questions (single answer: concepts, calculation etc)

- *Textbook Exercises: textbooks (Programming + Mining)*
- *Other Exercises: <https://www-users.cse.umn.edu/~kumar001/dmbook/sol.pdf>*
- *ChatGPT or other AI-based techs*